Linux Learning Lab — Lab 01: File Operations and Integrity Verification

Objective

Practice fundamental Linux filesystem operations and understand how to verify that a copied configuration file retains its original contents.

Environment: Ubuntu on WSL2
Certification track: Linux Foundation Certified System Administrator (LFCS)

Lab Scenario

A Linux administrator needs to create an application configuration file, preserve a copy and verify that the copy contains the same data.

Implementation

1. Create the configuration file

echo "server

_name=lab01" > configs/server.conf

echo "environment=development" >> configs/server.conf

2. Create a backup copy

cp configs/server.conf backups/server.conf.bak

3. Verify file contents

cmp configs/server.conf backups/server.conf.bak

echo $?

An exit code of 0 confirms that cmp found the files identical.

4. Calculate SHA-256 checksums

sha256sum configs/server.conf backups/server.conf.bak

Matching checksums provide strong evidence that both files contain identical data.

Technical Takeaways

- > overwrites file content; >> appends.
- cp copies files but does not independently verify backup integrity.
- cmp compares file contents byte by byte.
- sha256sum calculates a cryptographic checksum for integrity verification.
- A matching file copy is not necessarily a disaster-recovery-ready backup.
- Reliable backups require appropriate storage separation and tested restoration procedures.



